OUSSEMA AMRI

Full-Stack Software Engineer

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Summary

Full-Stack Software Engineer with experience delivering production systems in aerospace and internal business applications. Independently owned and refactored mission-critical engineering tools at a commercial rocket company, and developed an aerospace requirements-traceability SaaS (DO-178C / ISO 26262). Strong across Python, React, and TypeScript with hands-on AWS cloud architecture. AWS Certified Cloud Practitioner (CLF-C02).

Professional Experience

Rocket Factory Augsburg (RFA) *Augsburg, Germany*
Software Engineer – Working Student 06/2025 – Present

- Independently led the full refactoring and extension of two mission-critical internal tools used daily by Systems Engineering, Propulsion, Avionics, Launch, and Operations teams (∼42,000 lines across both codebases).
- Redesigned the Vehicle Parameters Database tool from a monolithic script into a layered FastAPI + PyQt5 architecture with a 5-router REST API, JWT auth, and 4-tier role-based access control; shipped 10 production releases.
- Cut tool startup from ∼90s to ∼6–8s via batch Git tree loading and 3-layer caching; eliminated O(n^2) loops with vectorized pandas for a 10–100× speedup on large schematics.
- Decomposed a 3,700-line legacy "god class" (zero tests) into 6 single-responsibility Python packages; built Git integration (pygit2) with delta-store commits, an auto-update system, CI/CD pipelines, and Windows installers.
- Also delivered an internal engineering document management system (FastAPI, React, PostgreSQL) with automated DOCX-to-PDF conversion, versioning, and approval workflows, deployed via Docker Compose and GitLab CI/CD.

Bouygues Telecom
Frontend Engineer Intern – Internship 06/2024 – 05/2025

- Developed and maintained internal business applications at Bouygues Telecom, working with the frontend team on the Wholesale and CDP back-office tools.
- Built React and TypeScript interfaces with reusable components and forms supporting business workflows.
- Maintained the CDP back-office, fixing defects and improving user journeys.
- Integrated Node.js APIs, including data validation, loading states and error handling.
- Contributed to Jest tests, Git code reviews and GitLab CI pipelines, with development tracked in Jira.
- Skills: React, TypeScript, JavaScript, REST APIs, Jest, Git, GitLab CI/CD and Jira.

Talan Tunisia *Tunis, Tunisia*
Software Engineering Intern – Full-Stack Jun–Aug 2023

- Developed HR platform features (profile management, document workflows, leave tracking, dashboards) using NestJS, React, PostgreSQL, Docker, and Jenkins CI/CD.

Personal Projects

Reqlume – Aerospace Requirements Traceability SaaS reqlume.com
Next.js 15, Django 5, PostgreSQL, Celery, Redis, TanStack Query, Playwright, Vercel, Railway

- Built a B2B SaaS replacing IBM DOORS / Jama Connect for aerospace teams managing requirements and compliance (DO-178C / ISO 26262); shipped 30+ features including approvals, change control, FMEA/RPN auto-calculation, and WeasyPrint compliance PDF export.
- Implemented a Polarion data import pipeline (CSV/Excel/ReqIF) with two-pass Celery tasks, tree building via django-treebeard, traceability graph with BFS cycle detection, and SSE real-time progress streaming.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
Frameworks: React, Next.js, Node.js, Vue.js, FastAPI, Django, NestJS, PyQt5, Celery
Cloud & DevOps: AWS (EC2, S3, ECR, CloudFront, Lambda, IAM), Docker, Docker Compose, GitLab CI/CD, GitHub Actions, Jenkins, Portainer, Grafana, Loki, Cloudflare
Databases: PostgreSQL, MySQL, Redis
AI & Tooling: OpenAI/Groq APIs, MCP Integration, Prompt Engineering, Claude Code, GitHub Copilot
Practices: Microservices, Event-Driven Architecture, REST/GraphQL API Design, JWT/RBAC, TDD, Agile/Scrum

Education

ESPRIT School of Engineering, Tunis – Diplôme d'Ingénieur in Computer Science (equiv. M.Sc., Bac+5) 09/2020 – 03/2026
Academic exchange (3+ semesters, ongoing): Philipps-Universität Marburg, Germany | 10/2024 – 03/2026

Certifications & Languages

AWS Certified Cloud Practitioner (CLF-C02) – Amazon Web Services
Languages: English (C1) | French (C1) | Arabic (C2) | German (A2)